

Tank Engines

The earliest tanks to see combat were powered by engines intended for large agricultural tractors (the British Mark I had a 105 hp Daimler-designed sleeve-valve 6-cylinder of 15.9 liters, which unfortunately belched smoke). Several interwar tanks used aircraft engines, such as the American V-12 Liberty, which powered the Mark VIII, BT-2, and BT-5, and early British Cruisers including the A13, Crusader, and Centaur. Other types of aero-engines of various configurations, often down rated, continued

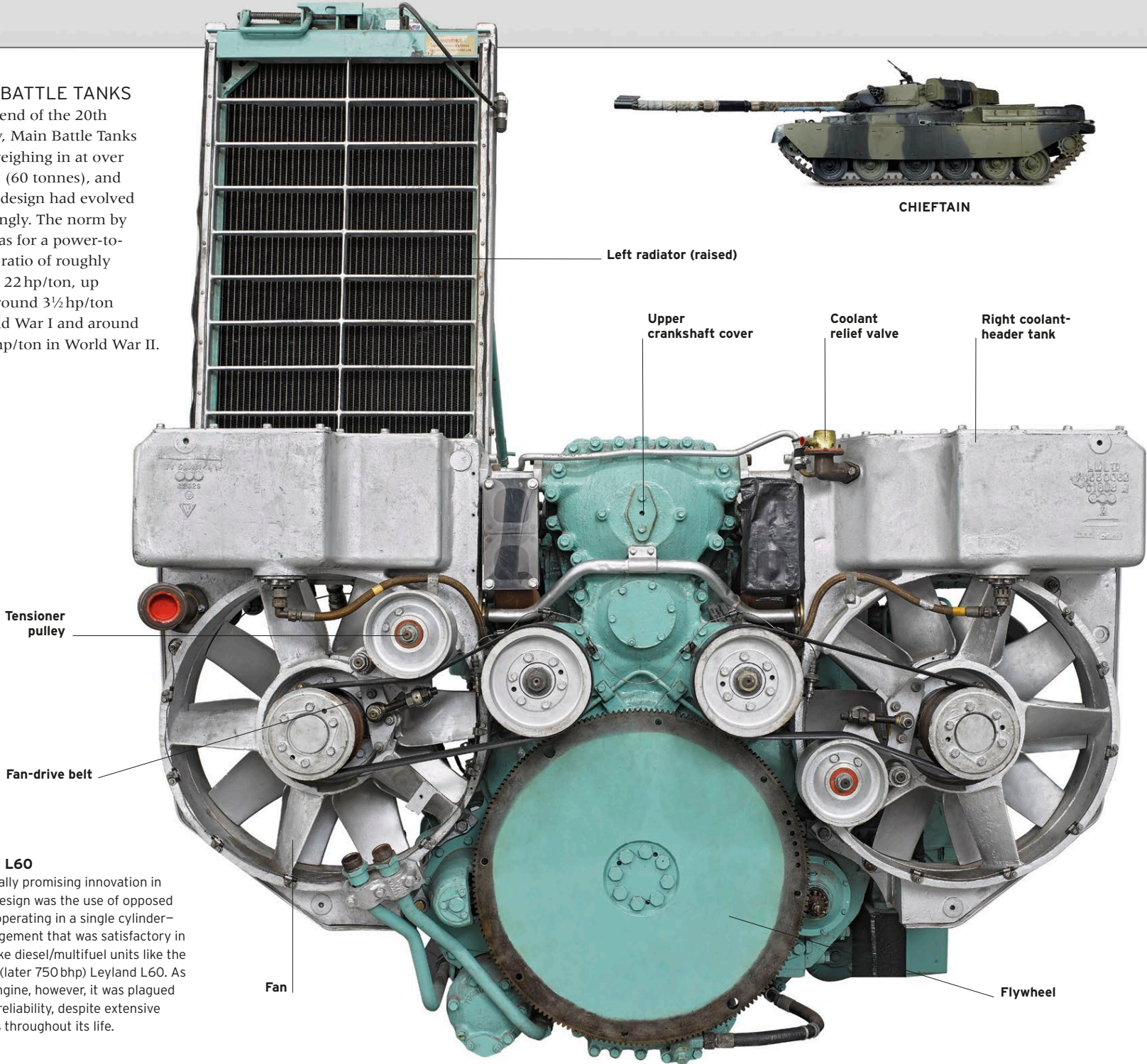
to power many Allied tanks throughout World War II, but already there was a move toward purpose-built units. By the 1950s, most tanks were propelled by 12-cylinder gasoline or diesel engines producing at least 750 bhp, many of which were air-cooled, and that de facto standard continued, with power output constantly being increased—even doubled—until well into the last quarter of the 20th century, when gas turbines first appeared, notably in the American M1 Abrams and the Soviet T-80.

MAIN BATTLE TANKS

By the end of the 20th century, Main Battle Tanks were weighing in at over 66 tons (60 tonnes), and engine design had evolved accordingly. The norm by then was for a power-to-weight ratio of roughly around 22 hp/ton, up from around 3½ hp/ton in World War I and around 11–14 hp/ton in World War II.



CHIEFTAIN



Leyland L60

One initially promising innovation in engine design was the use of opposed pistons operating in a single cylinder—an arrangement that was satisfactory in two-stroke diesel/multifuel units like the 695 bhp (later 750 bhp) Leyland L60. As a tank engine, however, it was plagued by poor reliability, despite extensive revisions throughout its life.